

Surendar UthamiNathan

Java Backend Engineer

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EDUCATION

ADHIYAMAAN COLLEGE OF ENGINEERING

BE IN COMPUTER SCIENCE

Grad. 2022 | First Class - 7.5 CGPA

SKILLS

Languages:

Java (8 / 11 / 17), JavaScript, SQL

Frameworks & Libraries:

Spring Boot, Spring MVC,
Spring Security, Spring Data JPA,
Hibernate, REST APIs

Tools & Infrastructure:

Maven, Git, GitHub,
Docker, Redis, Postman,
Spring Actuator, IntelliJ IDEA

Practices:

SOLID Principles,
Design Patterns (Builder, Factory,
Strategy, Singleton),
Microservices, CI/CD,
MVC Architecture

LINKS

LinkedIn:// Surendar-UthamiNathan

GitHub:// Surendar

COURSEWORK

Data Structures & Algorithms
Azure Fundamentals AZ-900 [Link]
Software Design & Architecture
Object-Oriented Programming
Operating Systems

ACHIEVEMENTS

Wipro TechFest Winner

2023

1st place among 500+ participants
in a competitive DSA contest in Java;
recognised by senior leadership.

Leetcode

Rating 1604 | Peak rank 3,809
350+ problems solved in Java.

EXPERIENCE

WIPRO TECHNOLOGIES | SOFTWARE ENGINEER

Jul 2023 -- Mar 2025 | Bengaluru, India

- Designed and delivered **5+ scalable RESTful services** using **Spring Boot, Spring Security, JPA, and MySQL**, supporting core business workflows with consistent uptime.
- Redesigned authentication with **stateless JWT + Spring Security 6**, eliminating server-side session state and reducing login latency by **~35%**.
- Resolved a high-concurrency data consistency bug using **ConcurrentHashMap** and fine-grained synchronization, cutting production incidents by **~70%**.
- Optimised **6 high-frequency JPA queries** with fetch joins and DTO projections, reducing average response time from **420 ms to 180 ms** (~57% improvement).
- Led **LLD design sessions** before each sprint, defining interface-driven contracts that reduced mid-cycle rework by **~40%**.

WIPRO TECHNOLOGIES | ASSOCIATE SOFTWARE ENGINEER

Aug 2022 -- Jun 2023 | Bengaluru, India

- Built **REST endpoints** for customer and billing workflows using **Spring Boot** with validation logic ensuring data integrity across services.
- Integrated **JWT authentication** and **Role-Based Access Control (RBAC)** using Spring Security, securing access to sensitive user data.
- Refined **Hibernate entity mappings** and HQL queries, improving query execution speed by **~25%** and reducing memory footprint for high-volume transactions.
- Authored **API and schema documentation**, accelerating new-developer onboarding and knowledge transfer.

PROJECTS

RATE LIMITER SERVICE | JAVA 17

- Implemented all **5 industry-standard algorithms** (Token Bucket, Leaky Bucket, Fixed Window, Sliding Window Log, Sliding Window Counter) behind a single **RateLimiter interface**, selectable at runtime via **Factory pattern**.
- Engineered **per-user ConcurrentHashMap isolation** with **synchronized** per-bucket critical sections; verified **zero token double-spend** under a 20-thread **CountDownLatch** concurrent stress test.
- Designed **RateLimitResult** response object carrying **remainingTokens** and **retryAfterMillis**, enabling production-grade **X-RateLimit-Remaining** and **Retry-After** HTTP headers.
- Applied **Builder, Factory, and Strategy** patterns in dedicated packages; immutable **RateLimiterConfig** validated at construction time prevents invalid runtime state.
- Architected **multi-profile RateLimiterService** supporting independent per-endpoint limits (login, search, payment) with config hot-reload capability.